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Instrumental measurements and sensory parameters

A. Hugi and E. Voirol, Nestlé Research Centre, Lausanne

2.1 Introduction

The human senses have always been used to assess food quality. For centuries they were the only instruments available, until scientific advances in physics, chemistry and biology, as well as the growing demands of the food processing industry, led to the development of analytical techniques aiming at understanding and controlling all aspects of food quality.

Although the senses of sight, hearing, taste, smell and touch are used daily in all aspects of our lives, their analytical application to evaluate food properties is relatively recent. Since this discipline is based on the perceptions and judgements of persons, with their inherent inter-individual and temporal variability, it is not surprising that the earliest contributions came from the fields of psychology and statistics. The nineteenth century saw the development of a scientific domain called psychophysics, concerned with the understanding of how people react to external stimuli. The focus was therefore less on the object being evaluated than on the subject evaluating it. Nevertheless, several current sensory test methods stem from that period (Peryam, 1990).

Modern sensory evaluation, defined as '... a scientific discipline used to evoke, measure, analyze and interpret reactions to those characteristics of foods and materials as they are perceived by the senses of sight, smell, taste, touch and hearing.' (Anon., 1975), really started to grow between the 1930s and the 1950s, with significant advances originating from governmental organisations, scientific or trade associations, and private companies (Peryam, 1990). Since then, sensory scientists have continued to develop new methods and to refine existing ones, but the food industry as a whole has been relatively slow to recognise the usefulness and validity of modern sensory methods. Over the last decades however, increasing competitive pressure has led food companies to pay more attention to consumer needs and preferences.

Today, foods from major suppliers usually offer comparable safety and nutritional value. Thus sensory characteristics are increasingly important for consumers as the differentiating factor between foods and brands. As a result, food companies recognise more and more the necessity to measure, optimise and control the sensory properties of

Table 2.1 Examples of practical questions answered by sensory tests

Is this reduction in viscosity perceivable?	Difference test
What is the effect of modifying the recipe (or the process) on a product's sensory properties?	Descriptive test
Can we substitute a specific ingredient by another one without affecting the sensory characteristics?	Difference test
What are the differences in sensory terms between our product and its competitors?	Descriptive test
What is the ideal sensory profile for this group of consumers?	Descriptive test combined with consumer test (preference mapping)
Is this product batch within specification?	Quality control test
How long can the product be stored before a significant change in sensory quality is noticeable?	Difference (or descriptive) test

their products. At the same time, the competitive pressure forces companies to master their cost structure, and to be more productive with fewer resources. Sensory evaluation is a good way to save money safely, for instance by shortening the development time of a new product or by reducing the amount of rework in a factory. This pressure on costs implies also a permanent search for efficient sensory methods delivering fast, business-oriented results.

Sensory evaluation offers today an array of analytical techniques with wide-ranging applications in research, market surveys, product development, and quality assurance. It is used to answer many practical questions arising in a food company, such as the examples shown in Table 2.1.

Given the high demand in time and resources of sensory evaluation, it is of interest to identify instrumental methods to replace it, particularly for routine quality control in a factory environment. Williams (1994) listed a vast array of chemical and physical information that can be related to sensory properties (Table 2.2). What is important to keep in mind is that a direct, causal relation between a physical/chemical measurement and a perceived sensory property cannot be simply assumed, it needs to be assessed and validated by a careful experimental study, carried on within the food product range of interest. While the concentration of sugar in a water solution has a univariate relationship with sweetness perception, this is probably no longer the case as the stimulus system becomes more complex, as in a confectionery bar, for example.

2.2 The nature of sensory perception

2.2.1 Senses as measuring instruments

Information from the outside world is received via our senses and processed by our brain to determine our reaction and behaviour in response to these signals. The sense organs, together with their related operational physiology, and neural signal processing systems, are evolved structures which were moulded by natural selection operating on environmental selective pressures (Stoddart, 1999). We are sensitive to two kinds of stimuli, physical (heat, pressure, acoustic waves, and electromagnetic radiation) and

Table 2.2 Chemical/physical measurements relating to sensory properties (from Williams, 1994)

Measurement	Related property
Light transmittance/reflectance	
• visual spectra	Colour
• tristimulus values	
• ultraviolet/near infra-red	Chemical composition, general quality
Size and shape	Appearance
• grading	General quality
• visual analysis	
Gas chromatography	
• volatile components	Aroma, flavour
Liquid chromatography	
• non-volatile components	Colour, taste, flavour
Simple chemical/physical measurements	
• pH, specific gravity, titratable acidity	Taste, general quality
• density, fibre content	Texture
• alcohol-insoluble solids	General quality
• moisture	Texture
• sulphur dioxide	General quality
Mechanical deformation	
• shear measurement	Texture
• compression measurement	
• puncture measurement	
Mastication	Texture
Ultrasonics	General quality
Molecular characteristics	
• bond length	Molecular aspects of aroma and taste
• spectroscopic information	
• stereochemistry	

chemical. The senses of sight, hearing and touch react to physical stimuli, while taste, smell and trigeminal receptors react to chemical stimuli.

2.2.2 Physical senses

Vision

The first contact we have with a food product is often visual. Colour and appearance guide us in the purchase of fresh products like fruits or vegetables, and food manufacturers pay due attention to the visual elements of packaging in order to appeal to consumers. In food processing and cooking, colour serves as a cue for the 'doneness' of foods and is correlated with changes in aroma and flavour (Lawless and Heymann, 1998).

The reader is referred to Chapter 3, by D. B. MacDougall, for detailed information on the physiological basis of colour vision and on the means to measure and communicate colour information. Beside colour, the assessment of food appearance includes many other visual characteristics that are related to physical properties: gloss, transparency, haziness, turbidity, uniformity of colour or surface, size and shape.

Hearing

This sense is of minor importance for the evaluation of foods, but it is not negligible: the noises of a cork popping out of a wine bottle or of a steak sizzling in the pan are signals

that we easily recognise and react to, even in absence of other sensory cues. Common noise characteristics of foods are crispiness, crunchiness and squeakiness (Meilgaard *et al*., 1999). In sensory terms, sounds are characterised by their pitch (frequency, Hz), intensity (loudness, dB) and persistence.

Physical vibrations in the air, in the approximate range of 16 Hz to 20 kHz, cause the eardrum to vibrate. The vibrations are transmitted via three distinctively shaped bones in the middle ear to create hydraulic motion in the fluid of the cochlea, in the inner ear. The cochlea is a spiral canal covered in hair cells of various lengths which, when agitated, send neural impulses to the brain. When we masticate foods, a large proportion of the sound travels by bone conduction through the teeth, jaws and bones directly to the cochlea (Vickers and Bourne, 1976).

Touch

The group of perceptions generally described as the sense of touch can be divided into 'somesthesis' (tactile sense, skinfeel) and 'kinaesthesis' (deep pressure sense or proprioception), both of which sense variations in physical pressure (Meilgaard *et al*., 1999). Various types of nerve endings in the skin are responsible for the somesthetic sensations we call touch, pressure, heat, cold, itching, and tickling. The particular sensitivity of the lips, tongue and hands enables us to perceive minute differences in the geometrical (gritty, chalky, fibrous, e.g.) and moisture (juicy, oily, greasy, e.g.) properties of foods.

Kinaesthesis is felt through nerve fibres buried in muscles, tendons and joints. We assess the mechanical properties of foods – hardness, cohesiveness, adhesiveness, denseness, springiness – via kinaesthetic perception of the stress exerted by muscles of the hand, jaw, or tongue, and of the resulting strain (compression, shear, rupture) within the food sample being handled or masticated.

2.2.3 Chemical senses

Olfaction

Olfactory receptors are located in a small patch of specialised epithelium in the roof of the nasal cavity. In this so-called olfactory epithelium, millions of neurones carry on one end several hair-like cilia immersed in mucus. Odour molecules bind to specialised receptor proteins within the ciliary membrane. It has been determined that mammals have around 1,000 different odour receptors, although they are capable of detecting at least 10,000 odours. Consequently, each different receptor must respond to several odour molecules and each odour must bind to several receptors.

At the other end of each neurone, a fibre known as an axon runs into the olfactory bulb of the brain. The neurones expressing a given receptor are randomly distributed throughout the epithelium, but project their axons to one or, at most, a few sites called glomeruli, in the olfactory bulb. As the positions of the glomeruli are topologically defined, the olfactory bulb provides a two-dimensional map that identifies which of the numerous receptors have been activated in the nose (Axel, 1995).

The dynamic range of olfactory receptors, between the threshold concentration and that producing saturation, is small compared to other senses like hearing and sight. On the other hand, the nose has enormous discriminating power: a trained perfumer can identify 150 to 200 different odour qualities. Moreover, olfactory receptors can be 10- to 100-fold more sensitive to some chemicals (e.g., allyl mercaptan: 6×10^7 molecules per ml of air) than the most sensitive gas chromatograph (ca. 10^9 molecules per ml) (Meilgaard *et al*., 1999).

Gustation

Taste receptor cells are clustered into taste buds, onion-shaped structures embedded within the lingual epithelium in the fungiform, foliate and circumvallate papillae, and found also in the soft palate and the epiglottis (Dulac, 2000). An adult has about 2,000 taste buds, about half of which are situated on the circumvallate papillae (Plattig, 1988). Each taste bud contains 30–50 cells that project small cilia into the salivary mucus covering the oral cavity.

We distinguish only five basic taste modalities: sweet, bitter, sour, salty, and umami (the specific taste elicited by monosodium glutamate), even though the molecular diversity of tastants is extreme, ranging from small ions to very large proteins. Thus, in contrast to smell, taste detects but does not discriminate between a large variety of molecules (Dulac, 2000). Because of this diversity, several pathways are thought to be involved in the reception and transduction of tastants, either via binding to receptor proteins (bitterness, sweetness, umami) or by interacting with ion channels in the taste receptor cell membrane (saltiness, acidity) (Laing and Jinks, 1996). The understanding of taste coding and information processing is still very limited, but the recent identification of a large family of 40–80 bitter taste receptors opens the way to significant progress on that topic (Adler *et al*., 2000).

Chemesthesis

In addition to taste and smell, there is a more generalised chemical sensitivity in the nose and mouth mediated by the trigeminal nerve, which is designated by the term chemesthesis (Green and Lawless, 1991). Comparatively less experimental effort has been expended in understanding the transduction processes involved in this sensory system, although the presence of several receptor mechanisms, ranging from receptor-specific to non-specific, has been hypothesised (Brand and Bryant, 1994).

A variety of flavour experiences arise from trigeminal stimulation: the fizzy tingle from carbon dioxide in soda, the burn from hot peppers, pungency from black pepper or other spices, the cooling effect of peppermint, or the nasal pungency of mustard and vinegar (Lawless and Heymann, 1998). Qualitative distinctions among chemesthetic sensations are not easy, either due to lack of vocabulary or to poor discriminative ability. Another confusing element is that some irritants, e.g. menthol, also have taste or odour properties, while some tastants or odourants, like salt or acetic acid, can produce irritation (Prescott, 1999).

2.2.4 Interactions

An important aspect of sensory perception in the context of this book is that foods generate a multitude of stimuli, which are processed more or less simultaneously to produce an integrated, overall impression. Along the way from the food item to the brain response, there are many possibilities for stimuli (particularly chemical ones) or sensory perceptions to interact, resulting in a response that is not simply the sum of individual responses to each independent stimulus.

Interactions between stimuli within the same modality have been extensively studied for tastants (see Breslin (1996) and Birch (1996) for reviews), and also for odourants. Inter-modality interactions have been reported, particularly between irritants and tastants, and between irritants and odourants (Lawless and Heymann, 1998). Whether these interactions are physiological (at the receptor site) or psychological (central processing) remains unclear. There is no evidence for a physiological interplay between taste and

olfaction (Laing and Jinks, 1996), but under normal conditions of real food consumption, there is certainly a cognitive interaction between taste and aroma (Noble, 1996), not to mention the additional influences of texture and appearance.

2.3 Sensory evaluation methods

2.3.1 Discriminative tests

Discriminative tests are used to determine whether or not a difference exists among samples. Two or several samples can be compared. Several types of discriminative test exist, each one corresponding to a particular situation. All are 'forced-choice' methods.

The **Triangle test** is probably the most universal sensory method. It is a non-directional difference test, used to detect an unspecified sensory difference between two samples. The panellist is presented with three coded samples: two samples are the same and one is different. He is asked to identify the odd sample, optionally he may comment on the intensity and type of difference. This test is useful in quality control, to check samples from different production batches, and to determine if any change of ingredient or process induces a detectable difference in the product, but it should be limited to relatively homogeneous products.

The **Duo-trio test** is also used to detect unspecified differences between two samples. Three samples are presented: one is labelled as reference and two are coded samples, one being the same as the reference and the other being different. The panellist is asked to identify which of the coded samples matches the reference. This test, similar to the triangle test in its objectives and uses, has the advantage of limiting the number of comparisons (i.e. it is well suited for intensely flavoured products) but it is less powerful due to a higher probability of selecting the correct sample by chance. The above two methods do not require any specific training of the panellists.

The **Pair comparison test** is a directional difference test, used to detect a specified sensory difference between two samples. The panellist is presented with a pair of coded samples and is asked to determine which one is more intense in a particular characteristic. This test is very simple, often used in quality control but its efficacy depends on the relevance of the chosen criterion of comparison.

The **Ranking test** is an extension of the pair comparison test. The panellist is presented simultaneously with a series of coded samples (3–10) and is asked to rank them in order of intensity for a single specific attribute. This test is generally used to screen samples from a large group. It does not give any indication about the magnitude of the difference but only an ordinal evaluation, which is why results from different ranking tests cannot be compared. When these methods are used to assess products, panellists should be selected according to their sensitivity to the attribute of interest. These two methods are also well suited to select and train panellists for descriptive tests.

Further to discriminative tests, the sensory analyst is generally interested in obtaining more detailed information about sensory characteristics of the products: descriptive tests aim to identify, measure and compare the sensory attributes of the products.

2.3.2 Descriptive tests

The aim of a descriptive test is to describe with a complete and non-redundant set of pertinent attributes the sensory properties of a product and to quantify the perceived intensity of each attribute on a scale. The outcome of this process is referred to as a

sensory profile. Several types of descriptive tests have been developed.

The **Flavor profile** was the first technique developed (Arthur D. Little Inc., 1958). A small panel of four to six trained assessors analyses and discusses the flavour characteristics in both qualitative and quantitative terms. The products are first analysed individually for aroma, flavour, mouthfeel, after-taste and overall impression or 'amplitude', then the assessors sit around a table and discuss their evaluation in group to reach a consensus. Extensive use of reference materials is made to achieve a complete description of the product. The final profile, consolidated by the panel leader, presents the detectable attributes ('character notes') in order of detection and their intensity scored on the following scale:

0 = not present
)(= threshold
1 or + = slight
2 or ++ = moderate
3 or +++ = strong

The final flavour profile being a group consensus, no statistical analysis can be carried out to compare different products.

The **Texture profile**® was developed in the 1960s by the General Food Research Center. It is defined as 'the sensory analysis of the texture complex of a food in terms of its mechanical, geometrical, fat and moisture characteristics, the degree of each present and the order in which they appear from first bite to complete mastication' (Brandt *et al.*, 1963). The Texture profile is rather similar to a flavour profile approach with panellists discussing results in order to reach a consensus. Each texture attribute is illustrated by references to represent each scale category. The number of scale categories can vary from five to nine, depending on the specific attribute. The reference foods are standardised with respect to brand name, handling procedure, sample size and temperature. The problem of such a method can be the variability of the references and the sensory fatigue due to extensive use of references during the test. Moreover, it is not realistic to separate texture from all other sensory properties because perceptions are interdependent and the exclusion of some other sensory attributes from the profile does not eliminate their perception.

The **Quantitative Descriptive Analysis**® (QDA) was developed by Stone *et al.* (1974) and is now widely accepted as the standard profiling method by most sensory analysts. It is a time-consuming and complex method but has broad applications such as description of complex products, comparison of products with identification and quantification of their differences, explanation of consumers' preferences, correlation with instrumental measures. Thus QDA is widely used in product development, quality assurance, shelf life testing, cost optimisation and market survey. Results can be related to instrumental, chemical or physical properties of the products.

The set up of this method requires several steps:

- select and train a panel
- generate the descriptive vocabulary
- prepare the questionnaire and the glossary
- create an experimental design taking into account the number of products, number of tasters and number of replicates
- organise the evaluation (sample coding, preparation, service, data acquisition)
- analyse the results and present them in a synthetic and understandable form.

The ideal number of panellists is 10 to 12. This number is large enough to avoid too much variability of the results dependent on too few subjects, and small enough to ensure an efficient and homogeneous training of the group. Prior to training, the panellists should be screened according to their familiarity with the type of product considered, to their ability to perceive differences, to verbalise and to work as a group.

After selection of the actual range of products to be profiled, the panellists are asked to generate a list of attributes to describe all the products' sensory properties, grouped by modalities: appearance, aroma, flavour, texture, mouthfeel and after-taste. Sometimes only some sensory aspects are considered, for instance colour and appearance of the product, in which case it is called Appearance Profile Analysis (APA) (Hutchings, 1994). The generation of vocabulary is facilitated by the panel leader who helps in reducing the number of attributes, reaching consensus on the meaning of terms and writing the definitions of the glossary. Usually 100 to 200 descriptors are generated by the panellists, then the vocabulary is reduced to 15–30 key attributes in the final questionnaire. The role of the panel leader is to facilitate the discussion and to organise sessions, but he must refrain from influencing the group. The panellists are then trained with product or ingredient references to illustrate the different sensory characteristics. This training helps them to identify and memorise the attributes and to develop a standardised evaluation procedure.

The setting up of the questionnaire includes the choice of a scale to rate the intensity of each attribute. The scale must be homogeneous for the whole questionnaire. Two main techniques are commonly used:

- Absolute profiling: the attributes are scored on a structured (numbered) or an unstructured (visual analogue) scale, generally anchored at both extremities by words. The scale is always organised so that the intensity increases from left to right. A 10-point scale or a 15-cm continuous/linear scale is most often used.
- Relative profiling: the attributes are scored in comparison to a reference sample situated in the middle of the scale. In this case the scale is organised so that the intensity goes from 'much less' on the left to 'much more' on the right. The relative scale is generally numbered from −5 to +5, with reference at 0. The relative scale is often used to assess deviations from the standard in QA tests or storage tests.

Training is essential to use the scale in a consistent and repeatable way. This is achieved by presenting several times standard samples illustrating the lowest and highest scores for each attribute, and intermediate levels. The complete training of a QDA panel is rather long; about ten 90-minutes training sessions are required to reach reliable and repeatable assessments. Nevertheless, and despite training, the panellists can have different perceptions and a certain amount of variability between them is unavoidable.

The experimental design has to take into account:

- the total number of products
- the number of products presented within a session (4 to 6, depending on the complexity of the products)
- the number of replicates (2 to 3)
- the number and availability of tasters
- the amount of time allowed to run the study.

To fulfil these requirements, products can be analysed according to a complete block design (all tasters analyse all products) or an incomplete block design (each taster analyses only some products). In any case the products must be evaluated independently and blind, with a monadic presentation (or against a reference for comparative profiling),

according to a balanced order of presentation to avoid bias due to order effect.

Statistical treatment requires a complete dataset. Univariate and multivariate methods can be applied to analyse and visualise results:

- The basic treatment consists of calculating the mean values and variability for each attribute and product.
- Statistical differences between products can be assessed by a two-way analysis of variance (ANOVA) on each attribute. This analysis, completed by a multiple comparison test (Duncan, Newman-Keuls, LSD...) allows the identification of significantly different products. Results are visualised graphically by bar charts or profiles (linear or spider webs) as shown on Fig. 2.1.
- In a second step, multivariate analysis such as Principal Components Analysis (PCA), Discriminant Analysis (DA) or Partial Least Square Regression (PLS) can be used to reduce the multidimensionality of the data and to look at the relationships between products and attributes. Results are visualised on a biplot graph. An example of PCA mapping of coffees is presented in Fig. 2.2.

Spectrum descriptive analysis^{TM} was developed from the flavour and texture profile methods (Meilgaard *et al*., 1999). In this case, the panellist scores individually the perceived intensities with reference to pre-learned 'absolute' intensity scales in order to obtain profiles universally understandable and usable by everyone. The method provides lexicons of attributes similar to those of QDA; each attribute is illustrated by standards that define a scale of intensity going from 0 to 15. The use of this method requires extensive training, because panellists have to be universal.

Free-Choice Profiling is an original technique of free description (Williams and Arnold, 1984). Panellists describe the products with their own words associated to their

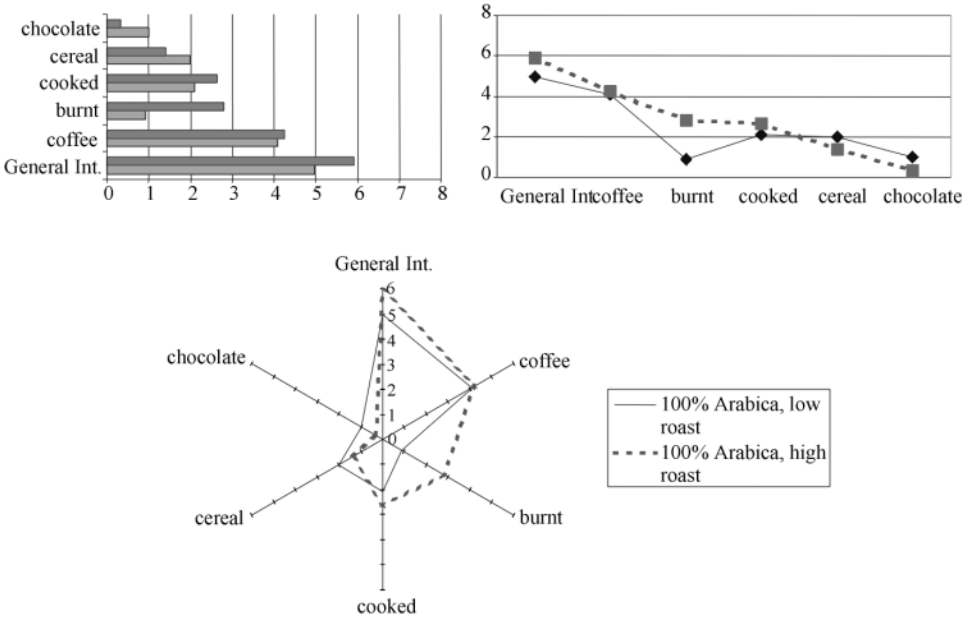


Fig. 2.1 Aroma profile of two different soluble coffees presented in bar chart, linear profile and spider web.

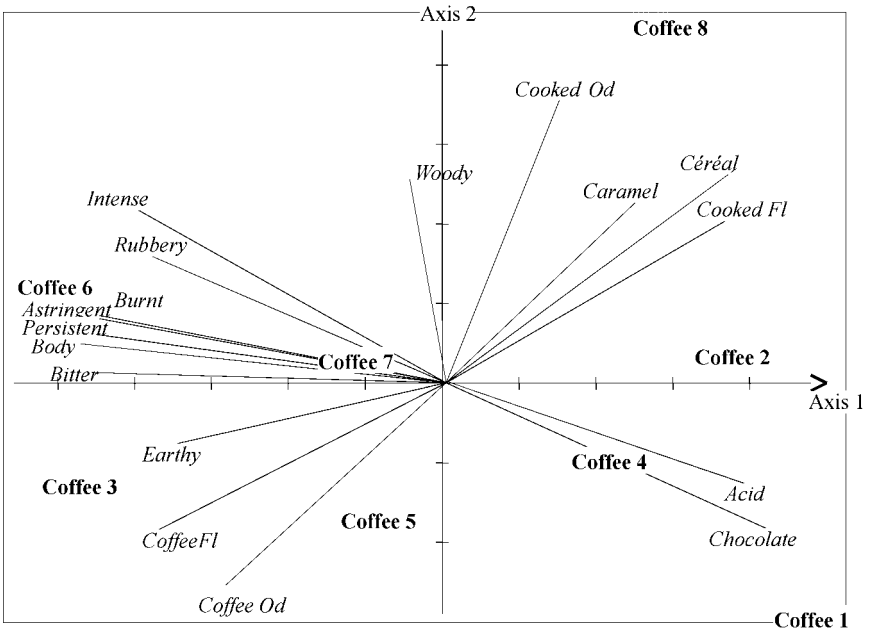


Fig. 2.2 PCA mapping of eight soluble coffees with varying Arabica/Robusta blend, and varying roasting level.

own scales. The advantage of such a technique is that it saves time since no specific screening or training is needed for the panellists. This method can be applied to all sensory aspects of the products or limited to some characteristics i.e. flavour or texture. Six to ten assessors are generally required. Data are analysed using a Generalised Procrustes Analysis. Free-Choice profiling has been often used in consumer tests, it can be considered as a good communication tool to relate marketing or consumer profiles to traditional QDA. It can be applied in panellist training, for descriptors consensus within a panel, or for agreement between several panels.

2.3.3 Quality control tests

The major application of sensory tests in production is to measure product conformity to a standard, in order to ensure quality consistency. A number of tests have been developed or adapted for quality control over the last 30 years, although some of the tests reported in the literature are not recommended for quality control purposes (Muñoz *et al*., 1992). Sound sensory tests for QC share some common and important features:

- product standards and specifications based on some consumer input, defining what the product is expected to be and the range of acceptable sensory variability
- a sensory panel trained on these specifications, using appropriate reference products
- a test methodology allowing quick and unambiguous decisions regarding product release or corrective actions.

The **Quality Ratings Method** is derived from the early sensory approach relying on ‘experts’ to judge the quality of products, and uses a panel to evaluate daily production on a quality scale ranging from ‘very poor’ to ‘excellent’ (or equivalent designations). Products below a certain quality level or grade are rejected. This method has enjoyed vast

popularity, even outside the factory environment, and several professional associations have published such methodologies for various types of products (e.g. for dairy products, see Bodyfelt *et al*. (1988)). Despite its popularity, this method presents several methodological flaws and is recommended only under special conditions (Muñoz *et al*., 1992).

Another widely used test is the **In/Out Method** (Nakayama and Wessman, 1979). A trained panel evaluates daily production as being either within ('in') or outside ('out') some well-defined sensory specifications. The percentage of panellists who score the product as 'in' determines the actions to take, according to decision rules set by the management. Advantages of this method are its simplicity, the direct use of the results and the panellists' motivation to participate in the decision-making process. Disadvantages are that relatively many panellists are needed to be able to analyse data in a statistical way, and the lack of information on the source of the problem when a product is 'out'. This can be alleviated by including comments or descriptive ratings in the test, but at the expense of its simplicity and speed.

The **Difference-from-Control Method** (Aust *et al*., 1985) involves rating on a scale how much different a sample is from a designated reference. Products are rejected if they are beyond a cut-off point on the scale. This method requires a consistent easy-to-hold or easy-to-reproduce control. Like the In/Out method, it is fast to implement and operate, and panel training is relatively easy. However, the test result indicates the amplitude of the difference, not its nature. Here too, adding to the test some attributes rating provides additional information, but that increases also the time needed to train the panel and to execute the test.

The most comprehensive approach is the **Descriptive Method** (Sidel *et al*., 1983). A well-trained sensory panel evaluates the intensity of several key sensory attributes, known to vary during production and to affect consumer acceptance. Specifications consist of tolerance ranges for each attribute, and a product falling outside any of these ranges is deemed unacceptable. This method provides the most detailed information about what is varying in a product, by how much and in which direction. Obviously, this is also the method requiring the most time for panel training and operation.

2.4 Sensory-instrumental relations

2.4.1 Principles

There are frequent occasions when it is desirable to relate sensory and instrumental data. This can be to understand the chemical and physical causes of the sensory properties in a product, or to reduce the sample load on a factory sensory panel by replacing some of the tests with instrumental analyses. Needs will differ in research and in production environments, as will the approaches and means used to address them. In any case, this will require a thoroughly planned and executed 'sensory-instrumental relation' study.

Definition of the objective

The first step is to define clearly and fully the objective of the relation study, i.e. which sensory property(ies) has to be related with which instrumental parameter(s) for which product (type), and to design the experiments to be conducted in view of this objective. This is particularly important with respect to the range and distribution of samples that will be considered (see below).

In a factory, a typical objective of such a study is to test instrumental methods that could be used to monitor product sensory quality (e.g., a micrometer for chocolate

grittiness, or a texture analyser for yoghurt firmness). Another goal can be to measure the physico-chemical characteristics of reference products used in sensory monitoring tests, to prevent them from drifting when they have to be renewed. A particular requirement in that environment is to employ affordable, robust instruments usable by unskilled personnel.

In a research context, relation studies are used to investigate the influence of the physico-chemical properties of a product or ingredient on its sensory properties, or to assess the effect of processing parameters (including storage) on them. This may require more sophisticated instruments, but on the other hand, robustness and operator training are not limiting factors. A few examples of research objectives are listed below:

- Assess the influence of process parameters (temperature, pressure, energy input, time, etc.) or equipment (size, supplier, etc.) on the sensory properties of a product.
- Optimise the processing conditions to achieve a given intensity of a sensory attribute, or to keep it within certain limits.
- Explore the potential of a different technology, or test new processing equipment.
- Find chemical 'markers' to follow the evolution of sensory properties with time (during process, during shelf life, under abuse).
- Identify physico-chemical characteristics that are related to key sensory attributes, in terms of consumer acceptance.
- Predict the final product sensory profile on the basis of the raw material chemical properties.

Experimental planning

Sensory considerations

The sample preparation conditions (temperature, type of water, etc.) must be standardised, and appropriate test parameters must be selected (scale, use of a reference, number of replicates, etc.). A balanced presentation design is necessary to eliminate first-position and carry-over effects. The performance level of the panel must be checked before the study. The trust in sensory data can be reinforced by showing its relationship with analytical data, but the panel performance cannot be assessed by any such relationship, particularly because the sample presentation design used for panellists assessment is different from the one used for samples evaluation. As for any sensory evaluation test, it is essential to ensure that samples are safe (microbiology, contaminants) before serving them to the panel.

Instrumental considerations

Forethought in method selection can considerably reduce the complexity of subsequent data analysis. The primary criterion for the instrumental method selection is its potential to be related to the sensory attributes identified as important. If several methods exist, only the most efficient should be selected. It is sometimes erroneously assumed that increasing the number of different characterisations reduces variability and increases the amount of information. More data is only potentially more information, but certainly more confusion (and more work). Adding more and more hay to a haystack will only increase the difficulty of looking for the needle in it. In statistical terms, this is referred to as the 'curse of dimensionality'.

Prior method validation is needed to ensure that the performance characteristics of the method meet the requirements of the intended analytical application. The potential of an analytical method can be reinforced by showing its relationship with sensory evaluation,

but its validity can not be assessed by any such relationship (likewise, the absence of such a relationship does not prove that it is not valid).

It is important to follow rigorous procedures during sample preparation and measurements. For instance, make sure the physical properties of interest (in relation to sensory properties) are not masked by the recent thermal or shear history of the sample.

Statistical considerations

It is generally easy to gather large datasets during factory production or product development, but most samples are not produced for the specific purpose of relating sensory data with instrumental characterisation. A sound experimental design is necessary to obtain a valid model for the sensory-instrumental relation. This implies the three following steps:

1. Identify the controllable parameters that will be varied (usually, these are compositional or processing parameters).
2. Define the experimental region by determining a range for each parameter that corresponds to its expected variation in real conditions.
3. Cover the defined region as homogeneously as possible, by using appropriate experimental design techniques.

The fundamental question in experimental design is not 'how many samples are needed?' but 'how homogeneously is the experimental region covered?' Moreover, it is crucial that the sample subsets used for sensory and instrumental characterisations are identical, which requires that samples be produced under stable and controlled processing conditions, and in sufficient quantity for each characterisation technique. Finally, data from any characterisation technique, even the best-mastered one, are never exact. Any measurement has some variability, the amplitude of which has to be known and controlled.

Data analysis

All available knowledge on individual datasets, and on possible relationships between different datasets, has to be gathered before analysing results. This helps to summarise data correctly and to select the most appropriate statistical methods.

Summarising data

Although some characterisation techniques (e.g. pH) yield a single value for each measurement, most techniques have a multiple outcome, i.e. they give a series of data points for each measurement. To simplify analysis and to focus on meaningful information, it is desirable to summarise multiple outcomes (Table 2.3). Sometimes pre-transformation of data has to be considered. For example, a chromatogram can be taken as it is or standardised according to the odour thresholds of known molecules.

Relating data

A relation is obtained mathematically by a model that expresses one dataset as a function of another dataset. If two sets of data are related by an existing and accepted deterministic model, no statistical method is needed. Unfortunately, deterministic models are generally not available and empirical model building must be applied, based on the actual observations. Purely empirical models are valid only within the explored experimental region and should not be extrapolated outside of it.

Data visualisation and descriptive methods, as well as prior knowledge, can provide hypotheses on the nature of the underlying relations. For instance, if a relation between

Table 2.3 Data summarising for different types of multiple outcomes

Dataset type	Definition	Summarisation
Signal	Characterisation of the product as a function of one variable (e.g. time). Examples: texture analyser, gas sensors.	A signal can often be summarised by a reduced number of parameters: maximum peak intensity, area under curve, asymptotic behaviour, perimeter of the curve (or the related fractal dimension).
Distribution	Distribution of the values of a single attribute in several classes. Example: particle size distribution.	A distribution can often be summarised by a reduced number of parameters: mean, mode, percentiles, standard deviation.
Profile	Characterisation of a product consisting of many single attributes. Examples: sensory profiles, chromatograms.	Often difficult to summarise because the characterisation is complete only when looking at all attributes together. However, some profiles may be composed of sub-profiles, like a sensory profile including appearance, odour, flavour and texture.

ice-cream texture and bubble size is investigated, the model is likely to contain the cube (volume) of the mean bubble radius. Knowledge-based models might have some predictive power outside the explored product range. The best approach with empirical model building is to start with a simple model and to increase its complexity only if it becomes more accurate, i.e. if the adjusted R^2 still increases. Finally, the chosen model should be validated. Validation consists of confronting the model – based on a given set of data (the so-called training set) – with another set of data (the validation set). It ensures that the model found not only applies to data from the current products, but also to data from any other product located in the previously defined experimental region. This check is essential whenever the aim of the model is to become a predictive tool.

Method transfer

If the sensory-analytical relation study aims at building knowledge on specific food products, ingredients or processes, its results will simply be applied within an existing project or prompt a new study. If however the purpose of the study is to set up an analytical method in production, additional steps must be taken to transfer the method to a factory. One frequent pitfall is that a relation study is made with very sophisticated instrumental methods that will never be reasonably transferred to the factory floor. If the objective of the study is to develop a quality-control tool (on-line or off-line), the constraints in terms of cost, complexity, and robustness must be considered in advance. Each quality-control method requires a reference measurement which frequently necessitates a more expensive, high-precision technique, which itself must be validated. This should be considered when establishing a method and its internal control plan.

2.4.2 Examples

Colour and gloss in sugar panned confectionery sweets

Together with colour, gloss is a key parameter of the aspect of sugar panned

confectionery sweets and an important factor driving acceptance for young children. For this reason, a study involving sensory analysis and instrumental measurement was conducted to investigate the effect of polishing on the visual appearance of this type of sweets.

Sugar panned confectionery sweets consist of a chocolate centre covered with several layers of sugar. The top layers are coloured, then covered with wax and polished in a drum to give a glossy, homogeneous and regular surface. The aim of this study was to optimise the coating process in order to reach the optimum gloss with the minimum polishing time. Different production batches of sweets were sampled directly from a production line every five minutes during the polishing process: from 0 (no polish, no wax) to 20 minutes polishing with wax. These samples were analysed by sensory methods and instrumental measurement with a spectrophotometer.

Sensory analysis

A panel of 13 trained assessors participated in the study. The colour tests took place in standardised conditions, i.e. in individual viewing booths (Macbeth Spectralight), lit with an artificial daylight source D65. The sweets were analysed using the Appearance Profile Analysis method. Five attributes of colour and appearance were scored on a 10-point scale: chroma, lightness, shininess, colour homogeneity, and surface regularity. As it is difficult to score the colorimetric attributes of colour due to a poor colour memory, the panellists were provided with colour references to anchor the scales of chroma and lightness attributes (colour chips from *Munsell Book of Color*).

A series of sweets of the same colour but with different polishing times were presented according to a randomised-balanced design to avoid bias due to presentation order. The same assessment was carried out independently for the different colours of sweets. Only the results of yellow and orange samples are reported here. Sensory analysis did not demonstrate any significant change in chroma nor degree of lightness with polishing time. However an important increase of shininess is observed at the beginning of the polishing process, i.e. up to about 10–12 minutes for the yellow and orange sweets. After this time the polishing process seems to become inefficient as no increase of shininess can be sensorially observed (Fig. 2.3). The homogeneity of the colour does not change with polishing, and the regular aspect of the surface is slightly increased up to 10 minutes polishing time.

Instrumental analysis

The colour measurement of sweets was performed with a ‘Macbeth Colour-eye 7000’ spectrophotometer equipped with a CIE standard daylight illumination source D65 (the same as for sensory analysis). A 3×8-mm view area was chosen. The presence of an adjustable specular port in this type of spectrophotometer allowed the measurement of the surface brilliance of samples by including or excluding the specular component (see Chapter 4 for more detail). Reflectance spectra obtained with specular component included (SCI) and specular component excluded (SCE) were different. Only the SCE mode showed differences due to gloss value. In fact, calculation showed that Lightness (L*) was the parameter most influenced by gloss effect, Lightness increased with gloss in SCE mode, corresponding to a lighter aspect of the colour, while no significant change in L* was measured in SCI mode. The difference between the SCI and SCE reflectance, i.e. ΔE^* (SCI − SCE), gave only the gloss value by excluding colour parameters. This gloss value could be related to polishing time and to the sensory evaluation of shininess.

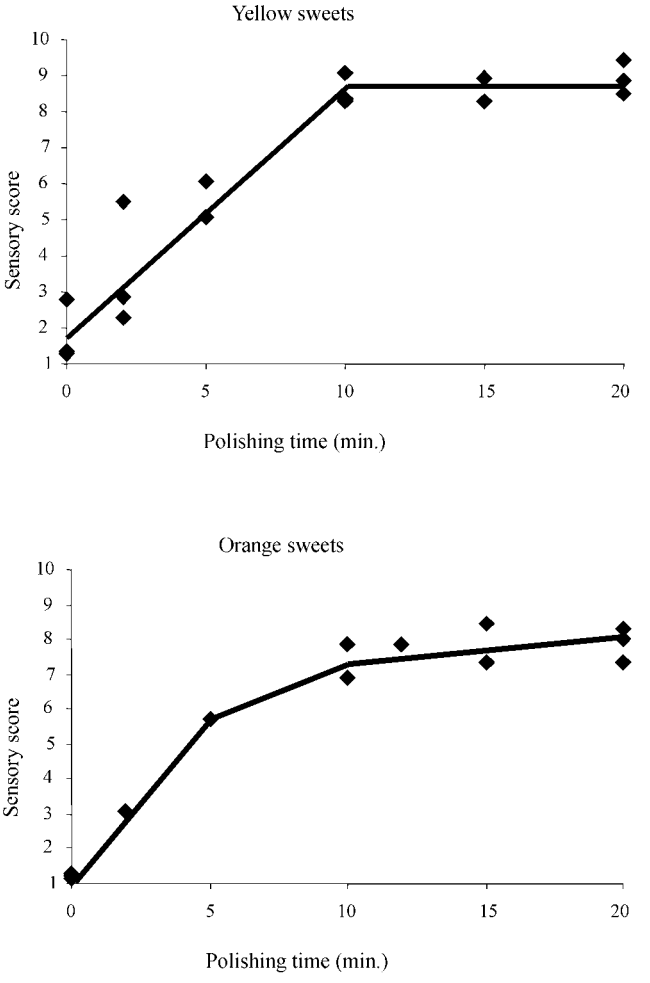


Fig. 2.3 Increase of shininess as measured by sensory analysis; evaluation on three different batches of yellow and orange sweets.

The colour measurements on sugar panned confectionery sweets were performed with specular component included and an observation angle of 10°. The results presented are the average of 10 measurements. In general, parameters of hue and lightness were not affected by polishing time. Chroma tended to decrease, indicating that the sweets became greyer or less vivid with longer polishing time (Fig. 2.4). This decrease was not visually perceived, but it suggests that there could be a risk of colour alteration if polishing time is too long. For each product colour, gloss increased significantly up to 10–15 minutes polishing time, less afterwards. The sensory scores of shininess were very well related to instrumental ones expressed by ΔE^* (SCI−SCE) (Fig. 2.5).

Conclusion

This study shows that most of the visual parameters of panned confectionery: hue, chroma, lightness and gloss can be easily and effectively controlled by instrumental measurements. These methods could be applied for quality control in factories.

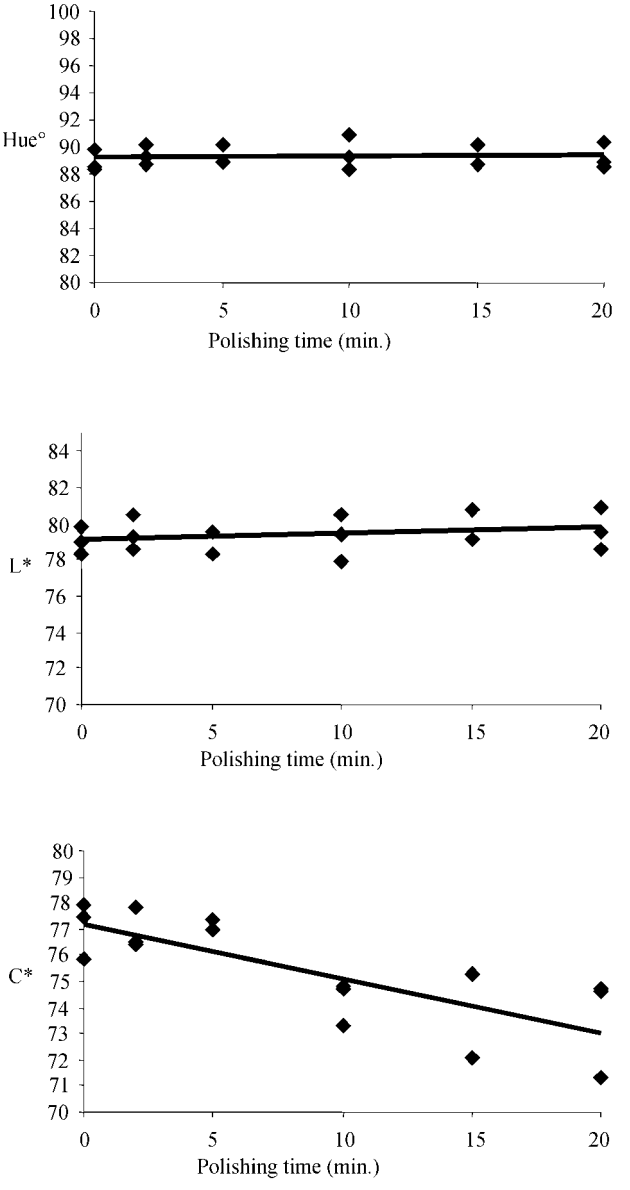


Fig. 2.4 Changes in instrumental hue, lightness and chroma with polishing time; measurement on three batches of yellow sweets.

However, some parameters of appearance such as colour homogeneity and surface regularity can only be examined by human vision, as they result from a multiplicity of stimuli synthesised by the brain as one piece of information. Sensory results, confirmed by instrumental ones, did not show any noticeable difference in colour and appearance between 15 and 20 minutes polishing, suggesting that the polishing time of this specific type of confectionery could be reduced to 15 minutes without affecting visual perception.

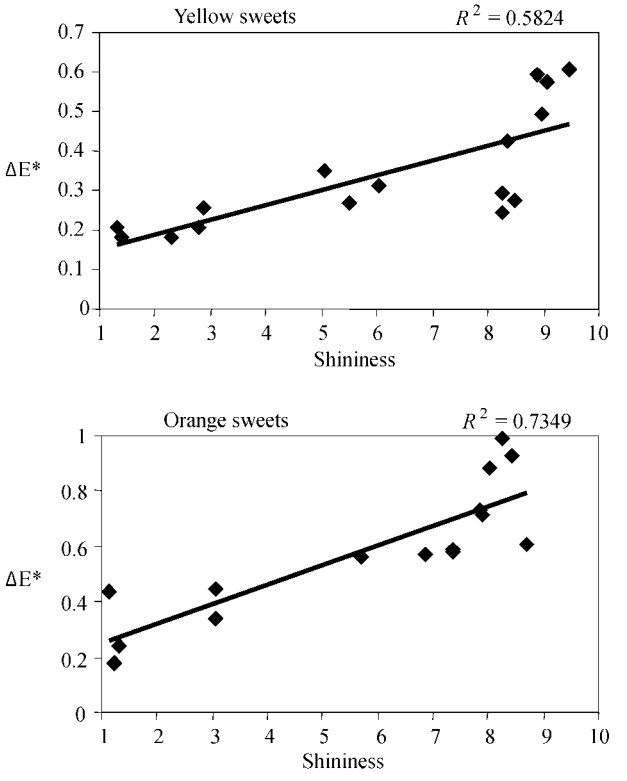


Fig. 2.5 Sensory-instrumental relationship for gloss; measurement on three batches of yellow and orange sweets.

Colour of UHT milk

It is well known that fat content influences the perceived colour of milk. Conversely it has been demonstrated by several authors that visual parameters of milk or dairy products are used as cues for the evaluation of fat content (Tuorila, 1986; Pangborn and Dunkley, 1964). As there is a direct relationship between fat content perception and consumer preference, a change in milk appearance can influence acceptance. It is therefore very important to have a precise control of colour when developing new milk products, in order to maximise consumer liking. A study relating sensory and instrumental analyses of milk colour was conducted, first, to determine the visual tolerance of milk at two different fat levels and secondly, to assess how the colour and appearance attributes of milk vary with the fat level.

The samples studied were prepared from two commercial UHT milks: skim milk (0% fat) and whole milk (3.8% fat). All intermediate fat levels were prepared by mixing the two milks in the relevant proportions. A panel of 12 assessors recruited according to their colour vision accuracy was trained for milk analysis using discriminating, ranking and scoring tests. The sensory assessments were performed in individual viewing booths lit with standard daylight source (D65, 300–830 nm, 6500K). The milks were presented in 30ml plastic cylindrical containers, filled up to the edges in order to have an easy view of the surface of the samples. The samples were observed according to standardised viewing geometry 0/45 (standard D1729-89 in ASTM, 1996) i.e., diffused incident light above the surface and viewing angle at 45°, with a distance of about 30 cm from the object. The

Table 2.4 Fat levels presented in the duo-trio tests, for REF 0% fat and 2% fat respectively

REF 0% fat	0.1%, 0.2%, 0.3%, 0.4%
REF 2% fat	For lower limit: 0.6%, 0.8%, 1.0%, 1.2%, 1.4%, 1.6% For upper limit: 2.4%, 2.6%, 2.8%, 3.0%, 3.2%, 3.4%

instrumental measurements were performed using a spectrophotometer Macbeth 580, with 10-mm sample depth, D65 illumination and SCE mode to take into account the effects of surface aspect that can influence sensory judgement.

Visual tolerance

The aim was to define the visual differentiation threshold of UHT milks having different fat levels. Two tolerance areas were measured for 0% and 2% fat. The limits of differentiation were determined by duo-trio tests using respectively 0% fat and 2% fat as reference samples. Table 2.4 indicates the samples presented. The differentiation thresholds were determined at the concentration for which 75% of panellists gave the correct answer to the duo-trio test. They were calculated from the linear regression between percentage of correct answers (transformed into Probits) and level of fat content (Fig. 2.6). The reflectance curves of all milks were measured simultaneously by reflectance spectrophotometry to obtain the ΔE* corresponding to visual limits of discrimination.

$$\Delta E^* \sqrt{\Delta L^{*2} + \Delta a^{*2} + \Delta b^{*2}}$$

The visual differentiation threshold for skim milk was calculated as 0.2% fat. This means that, up to 0.2% fat, milk is perceived as having the same colour as 0% fat milk. The corresponding ΔE* measured by spectrophotometry is 1.25ΔE* unit. Two

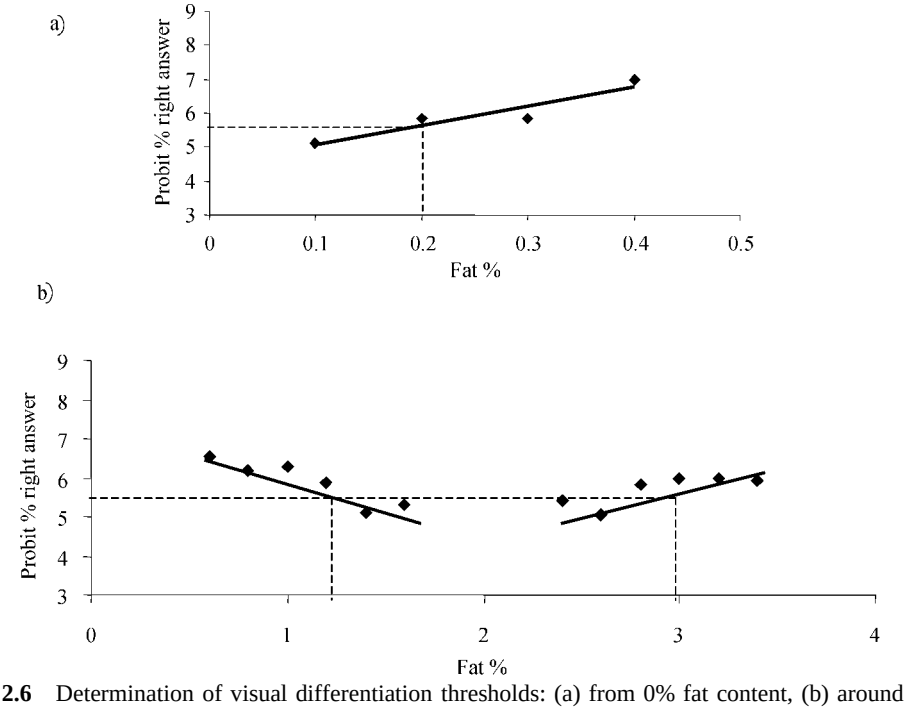


Fig. 2.6 Determination of visual differentiation thresholds: (a) from 0% fat content, (b) around 2% fat content.

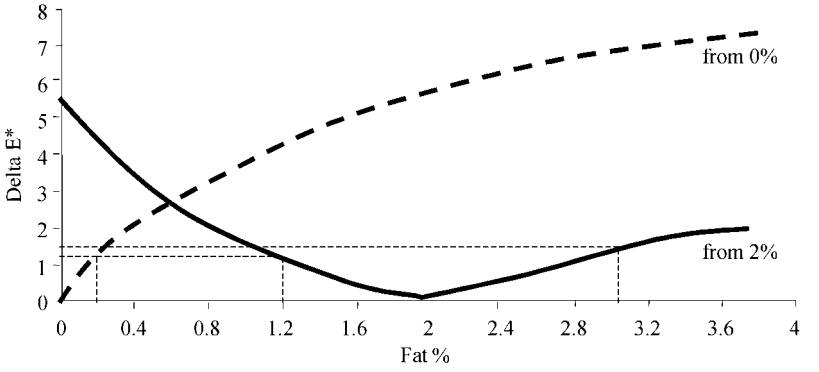


Fig. 2.7 Delta E* values as measured from both 0% fat and 2% fat standards.

differentiation thresholds were calculated for low-fat milk (2%): 1.2% (lower value) and 3% (upper value). This means that the colour of a milk with a fat content of between 1.2% and 3% will be perceived as that of a 2% milk. The corresponding ΔE*s measured by spectrophotometry are 1.46ΔE* units and 1.25ΔE* units, as shown in Fig. 2.7.

Sensory and instrumental measure of colour

Milks of varying fat content were profiled using the Appearance Profile Analysis method already described (see previous example). At the same time they were measured by spectrophotometry using standard methodology. Both methods showed that most colour and appearance attributes are affected by fat content. There was good visual discrimination for the three main colour attributes:

1. Hue changed from greenish to yellowish with increasing fat content. This change was particularly marked up to 2% fat.
2. Chroma or colour saturation decreased as fat content increased, i.e. the green colour of low-fat milks seems more saturated than the yellow colour of high-fat milks.
3. Lightness increased as fat content increased. This means that the green colour of low-fat milks appears darker than the yellow colour of high-fat milks.

The spectrophotometry measurements showed that the reflectance percentage (R%) increased across the whole visible range as fat content increased. This phenomenon is particularly marked at the higher end of the reflectance spectrum (Joshi, 2000). As discussed in Chapter 4, this is due to the higher scattering efficiency of fat droplets at longer wavelengths (Fig. 2.8).

Figure 2.9 presents in parallel the changes of colour parameters with increasing fat content, as measured by sensory analysis and by spectrophotometry. Changes were similar and the correlation between sensory and instrumental data was good, particularly for hue and lightness. The correlation was also meaningful for chroma ($R^2 = 0.88$), although results showed that increasing instrumental chroma corresponded to decreasing sensory chroma (Fig. 2.9c). This negative relationship can be explained by an increasing sensory perception of whiteness, whereas instrumental chroma measures the deviation from an 'achromatic white' towards colour, resulting in an increase in C* (instrumental indices exist to measure specifically whiteness, i.e. 'CIE Whiteness Index' mostly used in the paint industry).

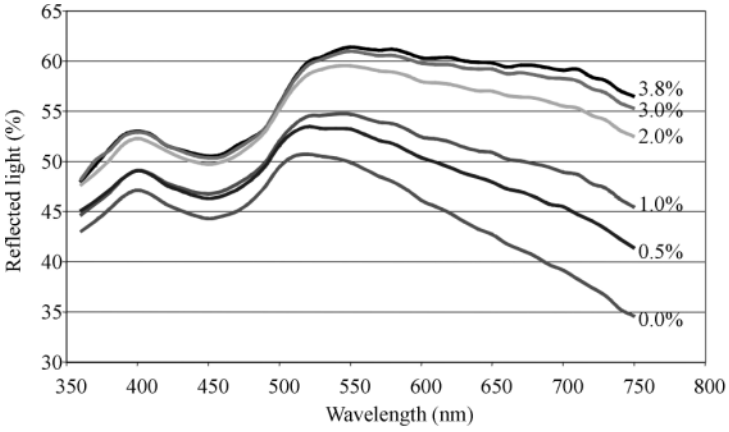


Fig. 2.8 Reflectance spectra of milks with varying fat content.

Conclusion

This study demonstrates that sensory analysis is generally a necessary prerequisite to define production standards. In this case, visual differences are determinant to fix tolerance area, and physical values (i.e. ΔE^*) can be determined in relation to sensory data. This principle can be extended to other types of perceptions such as texture, taste or flavour. The instrumental methods developed to assess colour of UHT milks are very well related to sensory ones, so they can be used to control and optimise the colour of new milk products.

Crispness of fried products

Crispness is a desirable quality in many food systems, such as breakfast cereals, confectionery inclusions, savoury snacks, fried products, and so on. Much effort has been spent in academia and industry to develop an instrumental method to measure crispness, in order to understand and improve the crispness of products, and to ensure consistent high quality in production. These developments involved instruments that mimic the senses used to gauge crispness, i.e. tactile and acoustic. Typically, tactile information is obtained from the analysis of force-deformation curves of compressed samples. The acoustic part is analysed from recordings of sounds produced either by actually biting and chewing the product or by mechanical imitation. The following example describes an acousto-mechanical study on the crispness of coated fish products.

Samples

Five fish finger samples were produced in a pilot coating line with various breadcrumb materials. They were fried in a deep-fat fryer for better reproducibility and colour homogeneity. In order to obtain a broad range of crispness levels, a frying protocol was set up with two oil temperature settings (145º and 180º) and two frying times (3 minutes 10 seconds and 4 minutes 30 seconds). A total of ten samples varying in breadcrumb type, frying temperature and time were produced.

Sensory evaluation

A trained panel of ten people who had been selected according to their sensory abilities performed the sensory assessment. The training period consisted of 12 sessions of 90 minutes. In the first phase, the panel generated a profiling questionnaire focusing on appearance and texture, whereby panellists agreed upon the definitions and the evaluation

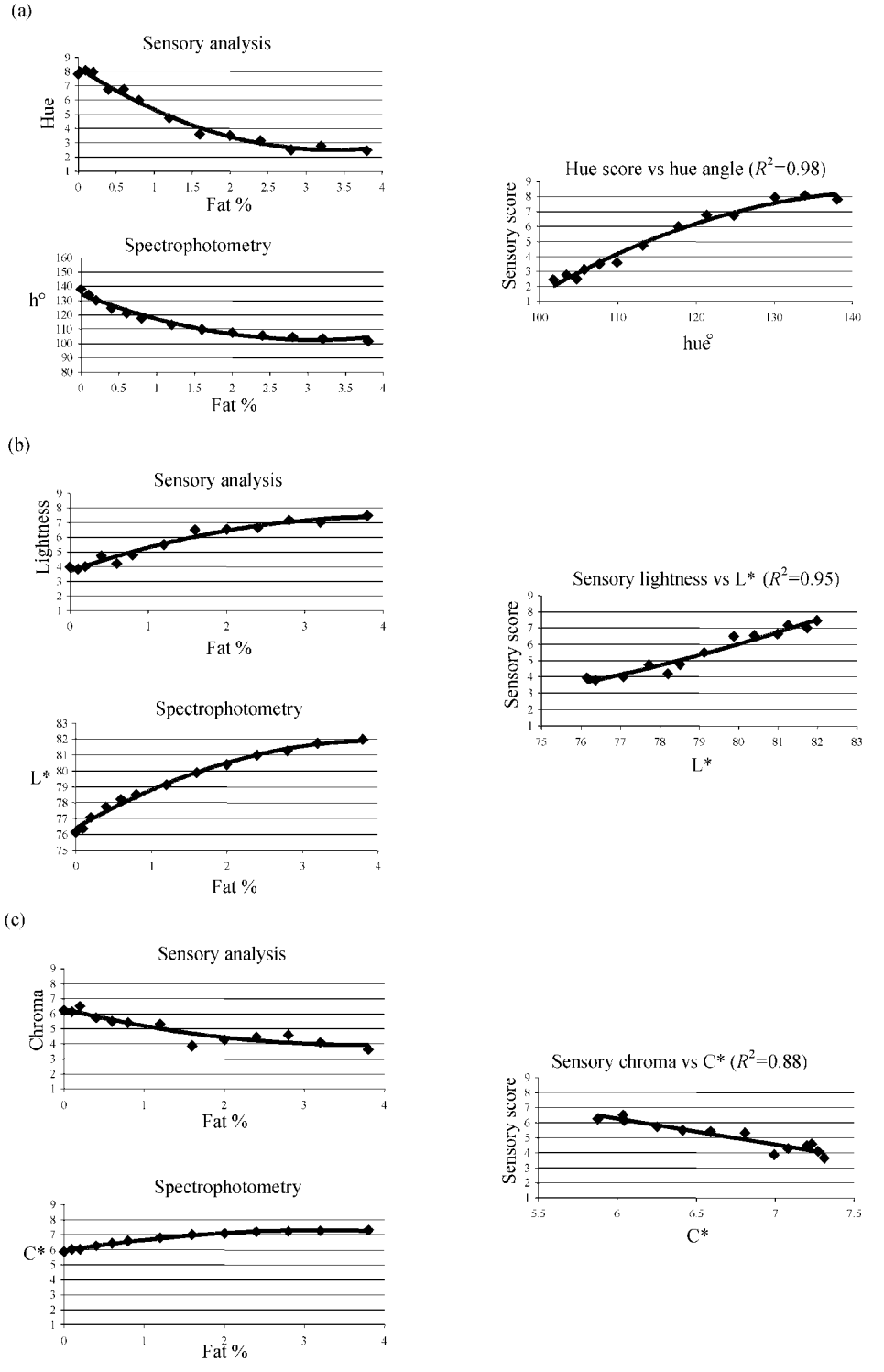


Fig. 2.9 Sensory and instrumental measurements of colour parameters: (a) hue (b) lightness (c) chroma.

procedures, and set the extremes of the scales. Then the panel was trained until its scoring was repeatable and consistent.

The questionnaire included 13 sensory attributes describing the appearance (swollen, golden, crumb size, crumb homogeneity, fatty), the texture in hand (crumbly), and the texture in mouth (breading: hard, crispy, fat; fish flesh: juicy, chewy, compact, stringy). Each attribute was scored on a 100-point continuous intensity scale anchored at both ends (not = 0; very = 100). All tasters evaluated products in three replications, according to a balanced experimental design with five samples per session. Samples were fried according to a rigorous procedure, kept in the basket for 30 seconds above the oil bath and served within 2 minutes to panellists, with randomised three-digit codes.

Analysis of variance showed that the products differed significantly ($p<0.05$) on all but one sensory attribute (*stringy fish flesh*). The difference was highly significant for crispness ($p<0.001$), with mean scores ranging from 17 to 79 on the 0–100 scale. Between-product significant differences were identified using Duncan's multiple range test.

Acousto-mechanic measurements

The mechanical properties of the samples were measured using a TA HDi Texture Analyser (SMS, UK). Two different sets of experiments were performed: in the first one, the fish finger was cut in the middle by a Kramer single blade to mimic the first bite with the incisors; in the second one, a five-blade shear cell was mounted to mimic molar mastication. Both experiments were run in 12 replicates.

The force-compression curves were first visualised, sample by sample, in the order of increasing sensory crispness (Fig. 2.10). This gave an overall impression of the results. In

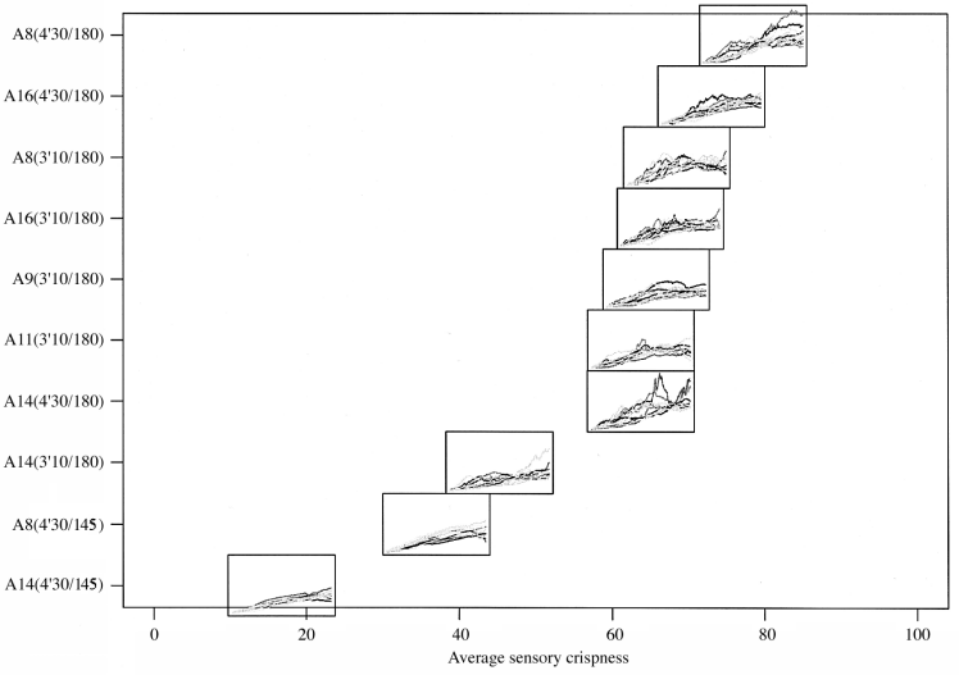


Fig. 2.10 Force-deformation curves for fish fingers, obtained using a Kramer single blade, plotted against average sensory crispness.

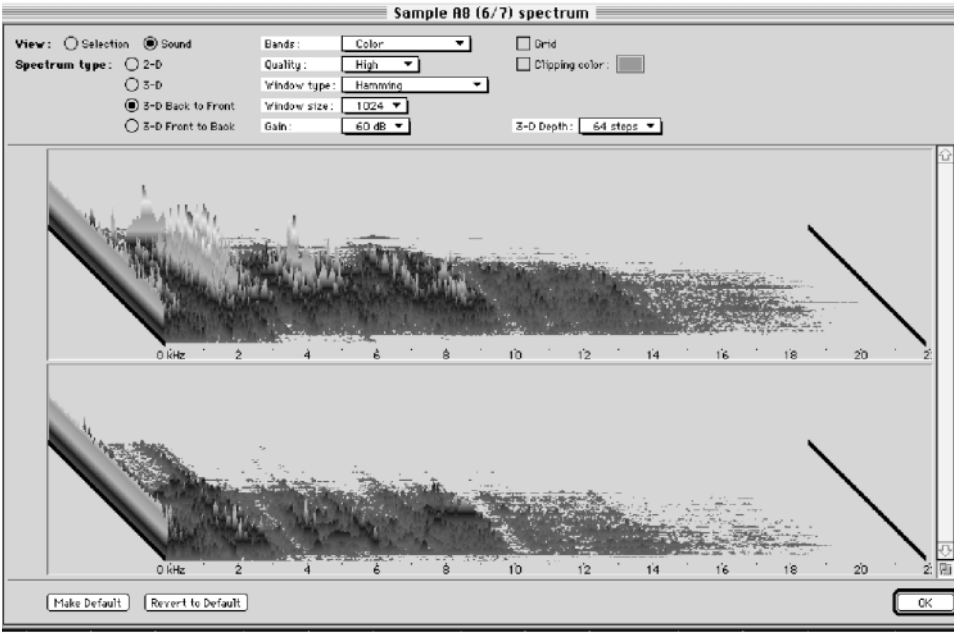


Fig. 2.11 Spectrogram of fish fingers fried for 4 minutes 30 seconds at different temperatures (top: 180ºC; bottom: 145ºC).

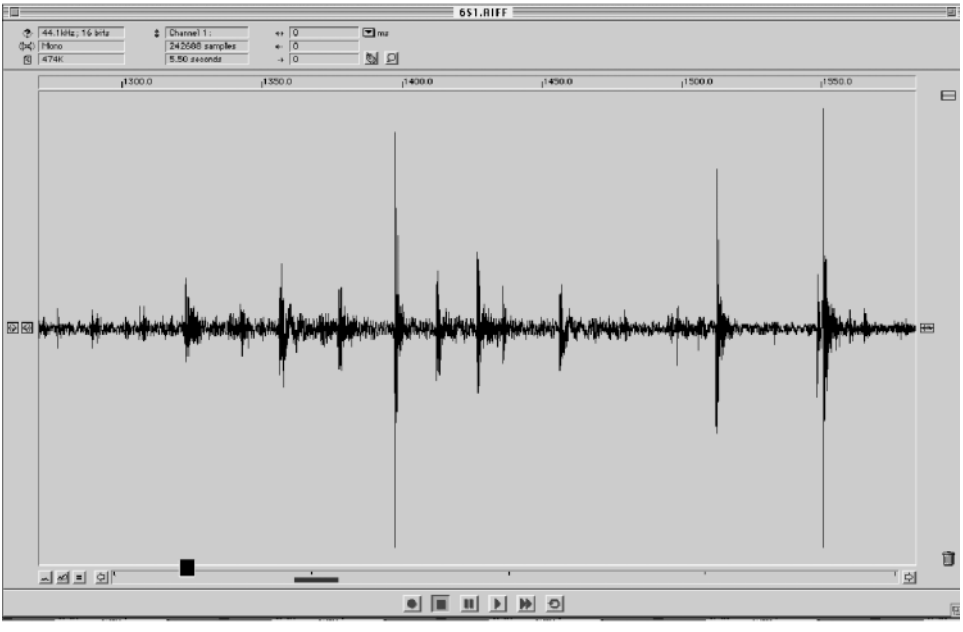


Fig. 2.12 A typical time waveform for a crispy fish finger.

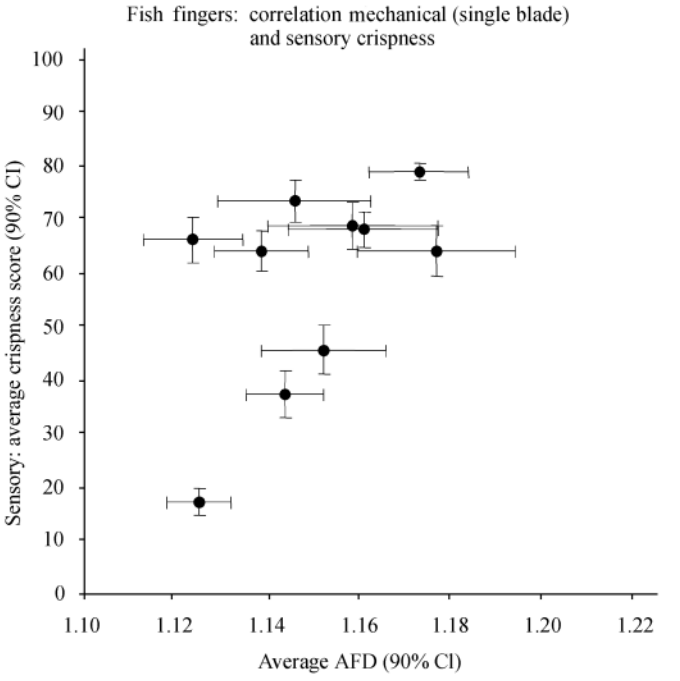


Fig. 2.13 Correlation between apparent fractal dimension (AFD) and sensory crispness. Mechanical measurements obtained using a single blade.

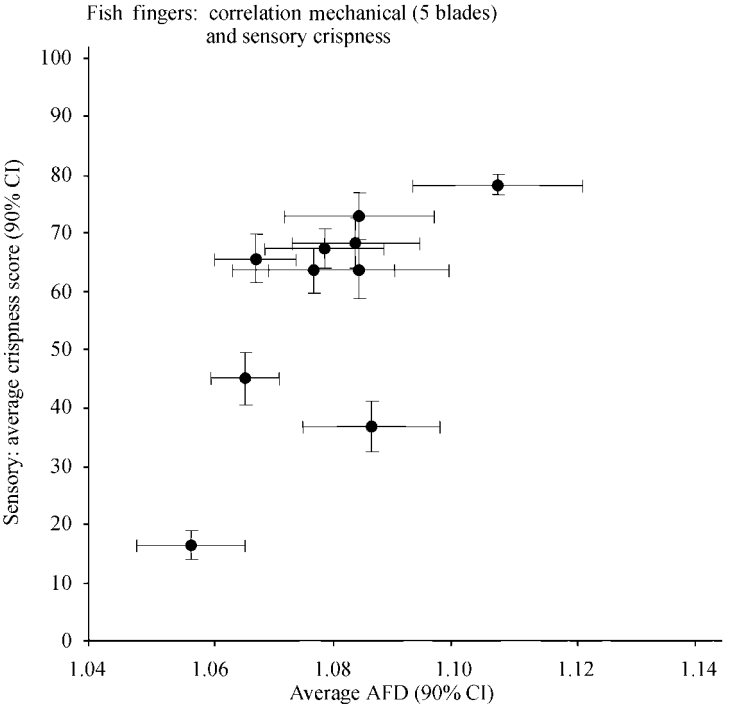


Fig. 2.14 Correlation between apparent fractal dimension (AFD) and sensory crispness. Mechanical measurements obtained using a five blade shear cell.

order to establish a quantitative relationship with sensory scores, the apparent fractal dimension (AFD) of the force-compression curves were computed using the Kolmogorov method, with the help of the *Fractals* software (Russ, 1994).

Whilst compressing the sample, the sound emission from the fish finger was recorded with a microphone, digitised using the sound sampling hardware of a personal computer (Power Macintosh 8600; acquisition software: Sound Maker 1.03) and stored directly to the hard disk. A Fast Fourier Transform algorithm was applied to the data, giving a three-dimensional spectrogram with time, frequency and intensity as axes (Fig. 2.11). These spectrograms are useful for a qualitative comparison of the samples. For a more quantitative evaluation, the waveforms in the time domain were analysed (Fig. 2.12). A typical crispy fish finger presents moments of low sound intensity interspersed with very short bursts of loud sound ('cracking events'). The number of such events above a predetermined sound level was thus added up for each experiment.

Results

The correlation between AFD and sensory crispness is shown in Fig. 2.13 (single blade) and Fig. 2.14 (five blade shear cell). Although a trend for a positive linear relationship can be observed, correlation coefficients are not satisfactory (single blade: $r = 0.52$; five blade shear cell: $r = 0.65$). Interestingly, the AFD for the single blade experiment correlates better with the attribute 'juicy' ($r = -0.7$), which seems to indicate that the texture of the fish flesh influences the force-compression curve with this attachment. The AFD for the five blade shear cell experiment correlates best with the attribute 'hard' ($r = 0.75$), which is highly correlated with 'crispy' for this set of samples. Thus this attachment provides a better indication of the coating texture properties.

The correlation between the logarithm of the number of cracking events and sensory crispness is shown in Fig. 2.15 (single blade) and Fig. 2.16 (five blade shear cell). The single blade experiments show an excellent correlation ($R^2 = 0.871$). For reasons discussed above, acoustic data correlate also very well with the attribute 'hard'.

This study illustrated the use of simultaneous mechanical and sound emission data to evaluate perceived crispness. In the case of fish fingers, acoustic data correlated well with the sensory perception, but the mechanical data (AFD) did not. The converse situation has been observed in other studies (with cereal extrudates, e.g.), where excellent correlation has been found with mechanical data and not with acoustic emission. This confirms earlier findings that our perception of crispness combines tactile and auditory sensations (Seymour and Hamann, 1988), but possibly in different proportions depending on the type of product.

Conclusion

Even the most promising, a priori, instrumental technique does not always relate well with sensory perception. A technique that works well for a given type of product is not necessarily valid for other product types. A combined acousto-mechanical approach offers the most versatility for measuring crispness in several product types.

2.5 Summary and outlook

This chapter briefly presented how our different sensory organs interact with food stimuli, and described the most commonly used sensory tests. Whether it is to determine if two samples are perceivably different, or to measure the nature and magnitude of

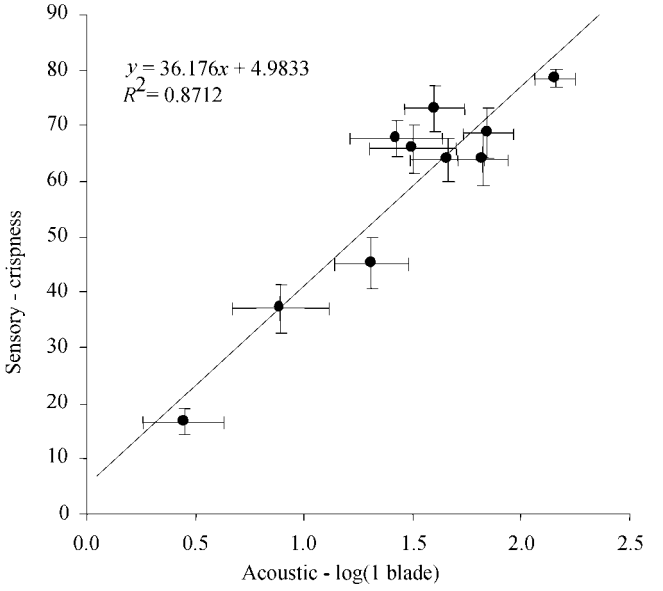


Fig. 2.15 Linear correlation between sensorially perceived crispness and acoustic emission data from a Kramer single blade experiment.

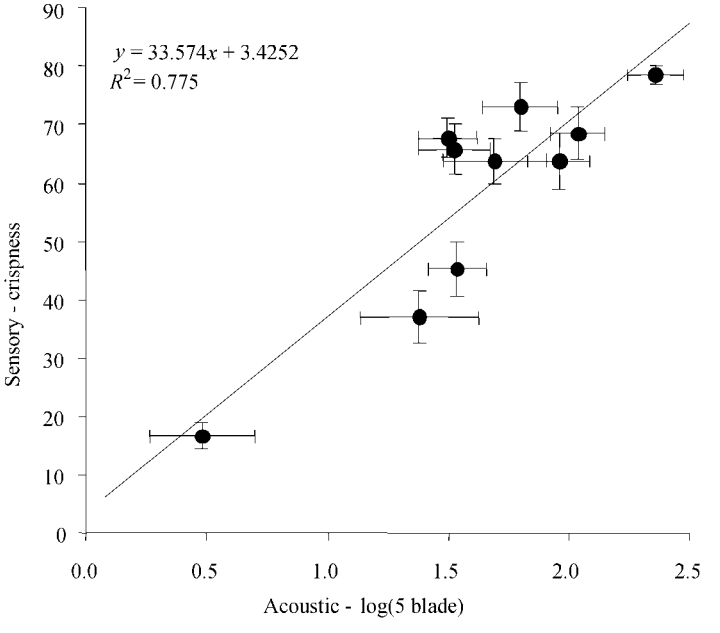


Fig. 2.16 Linear correlation between sensorially perceived crispness and acoustic emission data from the Kramer five blade experiment.

sensory differences among two or more products, or to assess the conformity of raw materials and finished products to sensory specifications, sensory evaluation plays a central role in the food industry. It remains the only way to evaluate how a food product is perceived by consumers, regardless of what the product actually is in physical or chemical terms. Clearly, consumers do not judge products solely on their sensory properties, and many other factors influence how they perceive foods: expectation, brand image, previous experiences, context, mood, socio-cultural background, and so on. Nevertheless, sensory quality is one element that a food company can control in order to satisfy and retain consumers.

Combining instrumental techniques with sensory evaluation is of great value in at least two areas of industrial food processing. On the one hand, it can be used to develop cheaper and faster instrumental alternatives to sensory tests for quality control, where assessors' availability, adequate facilities and sample throughput are often limiting factors. On the other hand, this combination can help to improve understanding of how ingredients, composition and processes influence food perception. In this chapter, we underlined the need for careful experimental and statistical methodologies in order to guarantee the validity of the outcome.

Sensory science is steadily progressing on several fronts; molecular biologists are characterising the receptor proteins in the gustative and olfactive receptor cells, neurologists identify the processing pathways of the sensory signals by brain-imaging techniques such as fMRI and magnetoencephalography, and experimental psychologists test the influences of memory, emotional context, familiarity and other external cues on sensory perception.

On the instrumental side, as well as electronic noses that are discussed later in this book, electronic tongues are now appearing, to measure solutes in liquids. New instrumental techniques are being developed to investigate further what happens during food consumption, either by simulating the mouth in terms of temperature, shearing, salivary environment (Roberts and Acree, 1995), or by direct sampling of volatiles in the nose, or solutes in the mouth, of a subject eating food (Taylor and Linforth, 1996), or by electromyographic recording of the movements of the jaw during chewing (Brown *et al*., 1998). This naturally leads to exploring the temporal dimension of sensory perception while the food item is processed in the mouth, to deal with physico-chemical aspects of foods such as flavour release, particle size reduction, melting process and interaction with saliva. Dynamic experiments use sensory methods such as time-intensity to track the evolution of a sensory characteristic over time, while time-resolved instrumental techniques monitor physical and/or chemical changes in the mouth and nose. This approach illustrates the complementarity of sensory and instrumental measurements in food research.

2.6 Acknowledgements

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